University of Engineering and Technology, Abbottabad Campus

LAB 01 – FALL 2026

Class:	5th semester Software Engineering	Date:	18 Aug 2026
Subject:	Software Construction	Instructor:	Engr.Rizwan Shah
Time Allowed:	3 hr 00 min.	Lab Engineer:	

Lab 1: Environment Setup, Version Control, and Introduction to Java

Objective: By the end of this lab, you will have a fully functioning Java development environment, a configured version control system, and your first working Software Configuration Item (SCI) hosted on GitHub.

Task 1: Install the Java Development Kit (JDK)

To write and run Java code, you need the JDK. It contains the Java compiler (`javac` for **Build-time**) and the Java Virtual Machine (JVM for **Run-time**).

Instructions:

1. Go to the official Oracle website or Adoptium (Eclipse Temurin) and download the latest Long-Term Support (LTS) version of the JDK (e.g., JDK 17 or 21) for your operating system.
2. Run the installer and follow the default prompts.
3. **Configure Environment Variables (Windows):**
    - Open the Start Menu and search for "Environment Variables".
    - Under System Variables, click **New** and create a variable named `JAVA_HOME`. Set the value to your JDK installation path (e.g., `C:\Program Files\Java\jdk-21`).
    - Find the `Path` variable, click **Edit**, and add a new entry: `%JAVA_HOME%\bin`.
4. **Verify Installation:**
    - Open a Command Prompt or Terminal.
    - Type `java -version` and press Enter. You should see the Java Run-time version.
    - Type `javac -version` and press Enter. You should see the Java Compiler version.

Task 2: Install the IDE (Apache NetBeans)

An Integrated Development Environment (IDE) automates the build process and organizes your workspace.

Instructions:

1. Navigate to the Apache NetBeans download page (netbeans.apache.org).
2. Download the latest installer for your operating system. *(Note: NetBeans requires the JDK to be installed first, which is why we did Task 1).*
3. Run the installer with the default settings.
4. Open Apache NetBeans to ensure it launches successfully.

Task 3: Create a GitHub Account & Install Git

In Software Construction, tracking the history of your code (Code Churn and Versioning) is mandatory. We use Git for local version control and GitHub for cloud hosting.

Instructions:

1. **Create an Account:** Go to github.com and sign up. It is highly recommended to use your university email address so you can access the GitHub Student Developer Pack later.
2. **Install Local Git:**
    - Go to git-scm.com and download the Git installer for your OS.
    - Run the installer using all the default settings (just keep clicking "Next").
3. **Configure Git:**
    - Open your Command Prompt or Terminal.
    - Tell Git who you are by running these two commands (use your actual name and GitHub email):
        - `git config --global user.name "Your First and Last Name"`
        - `git config --global user.email "your.email@example.com"`

Task 4: Create a Project in Java

We will now use NetBeans to generate a standard Java project structure using Maven.

Instructions:

1. Open NetBeans and go to **File > New Project**.
2. Under the **Categories** pane on the left, select **Java with Maven**. *(Maven is our Build Automation Tool).*
3. Under the **Projects** pane on the right, select **Java Application**. Click **Next**.

4. Configure your project:
    - **Project Name:** `Lab1_HelloWorld`
    - **Project Location:** Choose a dedicated folder for your university labs.
    - **Group Id:** `com.university.lab`
    - **Package:** `com.university.lab.helloworld`
5. Click **Finish**. NetBeans will generate the `src` and `target` folders discussed in the lecture.

Task 5: Write and Execute the Code

Let's create your first Java source code file.

Instructions:

1. In NetBeans, look at the **Projects** window on the left. Expand `Lab1_HelloWorld` > `Source Packages` > `com.university.lab.helloworld`.
2. Double-click the generated `.java` file (or right-click the package and choose **New** > **Java Class** if one wasn't created). Name it `HelloWorld`.
3. Type the following code exactly as it appears. *Notice that unlike C++ or Python, all Java code must exist inside a Class.*

```
package com.university.lab.helloworld;

public class HelloWorld {

    // This is the main entry point where the OS starts your program
    public static void main(String[] args) {

        System.out.println("Hello, Software Construction!");
        System.out.println("My Java Environment is successfully configured.");

    }
}
```

4. **Run the Code:** Click the green "Play" button in the top toolbar. You should see the text print out in the output console at the bottom of the screen.

Task 6: Version Control Your Project

Your code is currently a local **Software Configuration Item (SCI)**. We need to commit it to version control and push it to GitHub.

Instructions:

1. Log into GitHub on your web browser. Click the + icon in the top right and select **New repository**.
2. Name the repository `Software-Construction-Labs`. Leave it Public and click **Create repository**.

3. In NetBeans, go to the top menu bar and select **Team > Git > Initialize Repository**. Select your project's root folder.
4. **CRITICAL STEP - The `.gitignore` File:**
    - In your project folder, create a plain text file named `.gitignore`.
    - Add the word `target/` to this file and save it. This prevents Git from tracking your compiled Run-time binaries. We only version control Build-time source code!
5. In NetBeans, right-click your project name, go to **Git > Add**.
6. Right-click your project name again, go to **Git > Commit**. Type "Initial Lab 1 commit" in the message box and hit Commit.
7. Right-click your project name, go to **Git > Remote > Push**.
    - Paste the repository URL from GitHub into the Remote Repository URI field.
    - Enter your GitHub Username and Personal Access Token (or password).
    - Select the `master` or `main` branch and click Finish.